TUMU SANJAY

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ABOUT ME

MSc Advanced Computer Science student with a strong background in Full Stack Development and Artificial Intelligence. Experienced in designing and deploying scalable web applications using the MERN stack and building data-driven solutions using Python. Skilled in RESTful API development, authentication systems, database design, and responsive UI engineering. Passionate about developing secure, efficient, and user-centric software solutions aligned with modern industry standards.

TECHNICAL SKILLS

Programming
- Python, Java, JavaScript, C, C++, Java

Frontend Development
- HTML5, CSS3, React.js, Bootstrap, Responsive UI Design

Backend Development
- Node.js, Express.js, RESTful APIs, JWT Authentication

Databases
- MongoDB, MySQL, Excel from Data base

Tools & Platforms
- Git, GitHub, Postman, VS Code, NPM

Core Concepts
- Data Structures & Algorithms, Object-Oriented Programming, MVC Architecture, Software Development Life Cycle (SDLC), Agile Methodology

EDUCATION

University of Hertfordshire, Hatfield, United Kingdom
Msc Advanced Computer Science
2025 - Present

Relevant Coursework:
- Programming for Software Engineers, Artificial Life with Robotics, Computational Algorithms and Paradigms, Team Research and Development, Responsible Technology.

CMR Institute of Technology, Hyderabad, India
B.Tech in Artificial Intelligence & Machine Learning
2021 – 2025
CGPA: 7.2 / 10

Relevant Coursework:
- Machine Learning, Deep Learning, Artificial Intelligence, Probability & Statistics, Full Stack Web Development, Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

INTERNSHIP EXPERIENCE

CareerX Student Tribe
Hyderabad, India. Dec 2023 – Aug 2024. CareerX

- Delivered 8+ web development projects under team mentorship using React.js, HTML, CSS, and JavaScript.
- Collaborated with cross-functional teams of 5–7 members to build and deploy responsive client-based web applications.
- Worked with 3+ external organizations to gather requirements and deliver customized frontend solutions.
- Improved UI responsiveness and reduced page load time by approximately 20% through code optimization.
- Contributed to API integration and debugging, resolving 15+ frontend issues during development lifecycle.

(PR & Logistics Head) – StreetCause
Hyderabad, India. Jun 2023 – Jul 2024. StreetCause

- Led Public Relations and Logistics operations for social and technical events.
- Coordinated with sponsors, volunteers, and organizing committees.
- Managed event planning, budgeting, and execution strategies.
- Supervised teams to ensure structured workflow and timely completion of tasks.
- Strengthened leadership and problem-solving abilities in high-pressure environments.

Hyderabad Youth Association (HYA)
Hyderabad, India. Jun 2023 – Jul 2024. StreetCause

- Participated in community outreach and youth engagement initiatives.
- Coordinated volunteers for social awareness and development programs.
- Assisted in planning and executing organizational events.
- Enhanced communication, leadership, and organizational skills.

PROJECTS

MERN Stack E-Commerce Platform (OLX-Style Marketplace)
Tech: MongoDB, Express.js, React.js, Node.js
Link: E-Commerce

- Built full-stack marketplace supporting user authentication and product bidding for 100+ test users.
- Designed RESTful APIs handling 20+ endpoints.
- Implemented JWT authentication improving login security.
- Structured backend using MVC architecture reducing code redundancy by 30%.

Notes Application – React.js
Tech: React.js
Link: Notes

- Developed a dynamic CRUD-based note management system.
- Executed state management using React Hooks.
- Engineered responsive UI for cross-device compatibility.

Cryptocurrency Tracker App
Tech: Web development
Link: Crypto app

- Built interactive frontend displaying real-time cryptocurrency information.
- Optimised search and favorites feature using local storage.
- Focused on optimized UI rendering and clean UX structure.

Gym Management Portal
Tech: MERN
Link: GYM app

- Delivered membership and class scheduling interface.
- Integrated responsive frontend with user-focused navigation.

Machine Learning Project
Tech: Machine Learning, Python
Link: ML

- Built and Architected a Machine Learning model to solve a real-world classification problem.
- Performed data preprocessing including cleaning, feature selection, and transformation.
- Trained and evaluated models using Python-based ML libraries.
- Applied practical ML workflow including dataset splitting, training, validation, and testing.

Fake News Classification – Deep Learning Project
Tech: Machine Learning, Python, TensorFlow, LSTM, NLP

- Trained LSTM-based deep learning model on 20,000+ Kaggle news records.
- Achieved 92% classification accuracy using TensorFlow and hyperparameter tuning.
- Reduced overfitting by implementing dropout layers and validation monitoring.

Water Potability Data Analysis – Statistical & R Project
Tech: R, Statistical Testing, Data Visualization

- Analysed dataset of 3,000+ water samples using R.
- Applied Welch Two-Sample T-Test to evaluate statistical significance ($p < 0.05$).
- Generated 5+ data visualisations including boxplots and distribution charts.
- Identified key factors impacting water potability with statistical confidence.

CERTIFICATIONS

- Full Stack Web Development – Internshala and CareerX
- Problem Solving (Basics & Advanced) – HackerRank
- Python – Infosys Springboard

ACHIEVEMENTS

- Awarded as a best member in the team CareerX Student Tribe
- Successfully built and deployed multiple full-stack applications.
- Strong foundation in AI/ML and modern web development technologies.

PROFESSIONAL INTERESTS

- Exploring emerging technologies in Artificial Intelligence and Full Stack Development
- Participating in coding challenges and problem-solving platforms
- Building personal tech projects and experimenting with new frameworks
- Photography and digital content creation
- Community engagement and event leadership